

Towards a Technology that Improves Focus, Comprehension, and Retention in E-learning for Higher Education students with ADHD

Aishee Das
School of Informatics
The University of Edinburgh
Edinburgh, United Kingdom
aisheedas2003@gmail.com

Aurora Constantin
School of Informatics
The University of Edinburgh
Edinburgh, United Kingdom
aurora.constantin@ed.ac.uk

Gennaro Imperatore
Computer and Information Sciences
University of Strathclyde
Glasgow, United Kingdom
gennaro.imperatore@strath.ac.uk

Abstract

Students in Higher Education (HE) with Attention Deficit Hyperactivity Disorder (ADHD) often face persistent challenges in video-based e-learning, including difficulties with sustained attention, comprehension, and information retention. Existing lecture platforms typically offer limited embedded support, while external tools for focus and note-taking operate outside the learning context and rely heavily on self-regulation. This paper presents the design, development, and evaluation of a digital learning tool that integrates support features directly into the lecture interface to assist students with ADHD. Developed using a user-centered design methodology, the prototype includes chapter-based video segmentation, AI-generated summaries and quizzes, a context-aware chatbot, and visual scaffolds such as mind maps and annotated slides. The design was informed by the input of students with ADHD and experts in Education, Accessibility, and Human-Computer Interaction (HCI). A high-fidelity prototype has been developed and evaluated to assess usability, relevance, and its potential to support focus, comprehension, and retention. The findings suggest that embedding accessible adaptive features in lecture content can reduce cognitive load and improve engagement for students with ADHD. The paper concludes with implications for inclusive e-learning design and directions for future research.

CCS Concepts

• **Human-centered computing** → **Accessibility technologies.**

Keywords

Inclusive learning environments, ADHD, Assistive Technology, Design-based research, Educational technology for Higher Education

ACM Reference Format:

Aishee Das, Aurora Constantin, and Gennaro Imperatore. 2025. Towards a Technology that Improves Focus, Comprehension, and Retention in E-learning for Higher Education students with ADHD. In *The 27th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '25)*, October 26–29, 2025, Denver, CO, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3663547.3759705>

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ASSETS '25, Denver, CO, USA

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ACM ISBN 979-8-4007-0676-9/25/10

<https://doi.org/10.1145/3663547.3759705>

'25), October 26–29, 2025, Denver, CO, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3663547.3759705>

1 Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental condition characterised by inattention (i.e. difficulty with following instructions, focus and organisation), hyperactivity (i.e. excessive movement and talking) and impulsivity (i.e. acting without thinking) [2, 19]. It is typically diagnosed in childhood, but often continues into adulthood, where its symptoms can evolve [5, 7, 21]. As individuals mature, hyperactivity and impulsivity usually decline, while difficulties with inattention often persist and become more prominent [6]. These challenges can be impactful for higher education (HE) students, where there is an expectation of self-learning and time management [17]. The transition from structured schooling to self-managed learning in HE can be challenging for students with ADHD, often resulting in negative experiences, poor academic performance, and reduced motivation [31, 32].

These challenges are exacerbated in e-learning contexts. In HE institutions, course delivery has steadily shifted towards e-learning as a key mode [24]. Lecture capture technologies such as Echo360 [15] and Panopto [27] have become common in digital learning environments, providing students with access to recorded lectures. However, lecture recordings are often long and non-interactive, leading to difficulties for students with ADHD, such as maintaining attention and engaging with the course material [12, 14]. Previous work found that, for students without ADHD, when viewing content passively, attention can start to decline as soon as 10 to 15 minutes, and this decline in attention is even more prominent in students with ADHD [13]. Although platforms such as Echo360 and Panopto include accessibility features, such as playback speed controls and captions, these platforms do not support engagement or help maintain attention. Additionally, they do not break down content into shorter and more manageable segments, a method that has been shown to support attention retention on platforms such as YouTube [22].

Several tools and strategies have been developed to help students manage attentional difficulties. For example, Glean [20] allows students to capture and annotate lecture content. Other tools and strategies include: the Pomodoro technique [8], and distraction blockers such as Cold Turkey [10], or Freedom [18], the aim of which is to reduce environmental distractions. However, these tools are not designed to be integrated into lectures and are heavily reliant on self-regulation and motivation of the students. They are not designed to adapt to moment-to-moment changes in attention,

nor do they provide interactive elements within the lecture material. Therefore, while these tools can be helpful, they often cannot support the sustained engagement and retention of information that students with ADHD require.

This work addresses this gap by presenting a unified digital learning tool that embeds multiple support features directly within the lecture interface. Unlike existing platforms, which tend to focus on a single function or rely heavily on self-regulation, our prototype integrates video segmentation, AI-generated summaries and quizzes, mind maps, a chatbot, and motivational elements into a single interface. This approach reduces cognitive load from switching between tools and aims to improve attention, comprehension, and retention for HE students with ADHD.

2 System Design

This tool was designed using a User-Centered Design approach [1].

2.1 Pre-design Studies

Semi-structured interviews (approximately 50 minutes each) were conducted with five university students (S1-S5) with ADHD from various academic disciplines in different UK universities. In the interviews, the participants discussed their experiences with lecture recordings, attention-related challenges, and the desired features of supportive digital tools.

A bottom-up (inductive) thematic analysis [9] was conducted to allow key themes to emerge directly from the data. The following themes were identified: attention-related challenges, distractions, existing adaptive strategies, and preferred tool functionalities. The participants described significant difficulties in maintaining attention during extended lecture recordings. One participant, S1, stated, *“If there’s a 50-minute lecture or five 10-minute videos, the shorter videos are a lot easier. . . I lose focus often thinking, ‘oh God, there’s still half an hour of this.’”* Similarly, participant S3 shared, *“Even if I’m interested, it doesn’t matter. I zone out inevitably.”* External distractions, such as notifications and background clutter, exacerbated these problems. Although students used adaptive strategies such as modifying playback speed, enabling captions, and setting self-directed goals, these were often insufficient in supporting sustained comprehension.

The participants were then asked to identify several desired features for a digital learning tool to support their study needs. Potential features identified included segmentation of long videos, generation of topic-based chapters, generation of summaries for each segment, embedding interactive quizzes to support understanding, and offering visual aids, such as mind maps, to show the linkage between lecture concepts. Participants also expressed interest in gamified elements to improve motivation, as articulated by S3, *“Gamifying the lecture, like watering a plant or building something after completing a segment, would motivate me.”* The use of gamified elements in technology-based tools for higher education students has also been shown to be beneficial, supporting engagement, motivation, and sustained participation in learning activities [3]. Across all responses, there was a strong emphasis on the importance of a minimal and intuitive interface to reduce cognitive load and support ease of use.

2.2 Low fidelity prototype

Based on the findings from the pre-design studies, a low-fidelity prototype was developed in Figma [16], prioritising simplicity, clarity, and support for ADHD-related needs. The prototype featured three sections: Course Library, a Lecture List, and Lecture Content. The Lecture Content section was composed of a segmented video player with visual chapter markers, concise chapter summaries, interactive quizzes, an AI chatbot, a Pomodoro timer, amongst other tools such as mind-maps, a note-taking feature, and slide annotation tools.

The interface design was informed by a combination of ADHD-oriented and HCI design principles. Lorna McKnight’s ADHD guidelines informed decisions such as the use of a calm, neutral colour palette to reduce visual overstimulation, minimalist layouts to support concentration, orderly grouping of related content, and concise and clearly signposted instructions [23]. These design decisions were complemented by Nielsen’s usability heuristics, which informed the layout and interaction design through consistent feedback, minimalist structure, intuitive navigation, and flexible user control [25].

2.3 Formative evaluation

Formative evaluation involved think-aloud sessions with the same participants who took part in the pre-design studies: five students and five experts (E1-E5) in HCI, Education, Accessibility, and Software Engineering. Students described the interface as *“very straightforward, very intuitive”* (S5), and appreciated the consolidated layout, *“Everything’s labelled and you’ve got symbols and pictures, which make it a lot easier”* (S4). Experts emphasised that *“having everything within the same platform makes sure your attention is a little more focused at that point”* (E3). Suggestions for improvement included clearer labelling of summaries, chapter-specific progress indicators, repositioning transcripts for better usability, and integrating a brief onboarding tutorial. The prototype was improved further based on the feedback to better match user needs and expectations.

3 System Implementation

The high-fidelity prototype was developed to improve the effectiveness of e-learning for HE students with ADHD (Figure 1). One of its core features is the segmented video player, which divides lectures into manageable, interactive chapters. Clickable sections on a video progress bar allow students to navigate generated chapters, helping to reduce attentional fatigue. Chapter summaries and quizzes were created using a GPT-3.5-based generative AI model via the Edinburgh Language Model platform [26]. The quizzes provide immediate visual feedback, reinforcing learning through active recall, a crucial strategy for information retention [11]. AI mind maps are also generated to visually represent concepts discussed in lectures with the aim of allowing students to structure and revise information. An integrated AI-powered chatbot feature supports immediate clarification of lecture content, significantly improving user autonomy and comprehension.

The prototype was designed in line with Web Content Accessibility Guidelines (WCAG) 2.2 [29], supporting high-contrast colour schemes (verified via WebAIM [30]) and full keyboard navigation to meet various accessibility needs. A built-in text-to-speech (TTS)

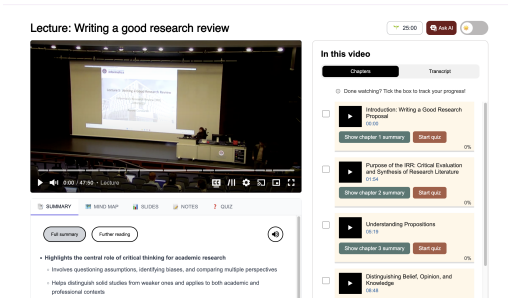


Figure 1: Main lecture interface.

feature, suggested by participants, enables audio playback of summaries, supporting multimodal engagement.

Additional tools include an annotation interface for personalising lecture slides and a structured notes section, both of which directly respond to participant suggestions. Reflective quizzes encourage active learning and support the development of essential self-regulation skills for students with ADHD. The application includes a Pomodoro timer with customisable focus durations and breaks to promote sustained attention and structured study habits. Students are rewarded for completing timed sessions by the appearance of visual rewards, which include animations and character badges that grow over time, to positively reinforce self-motivation and consistency in study routines. An onboarding tutorial, created with React Tour [28], guides users through the platform to help them feel at ease from the start.

4 Evaluation

The high-fidelity prototype was evaluated with seven university students diagnosed with ADHD (S1–S7) and five experts (E1, E2, E5, and two additional experts) in HCI, Accessibility, and Education. It involved cooperative think-aloud protocols, the System Usability Scale (SUS), and semi-structured interviews. A top-down (deductive) thematic analysis [9] was conducted using the following predefined themes: (1) Usability, (2) Attention Support, (3) Comprehension, (4) Information Retention, and (5) Accessibility.

Usability. The participants found the prototype highly intuitive, structured, and efficient. S4 highlighted the onboarding tutorial, noting, "I like the tour, how it jumps and highlights what it's talking about." S6 appreciated the clear chapter segmentation and remarked, "Nicely and clearly marked, split into decent chunks." The integrated design simplified workflow, reducing cognitive load. S1 said, "Usually I'd have to do these things separately. It's really cool that everything's integrated." Usability received SUS scores averaging 94 (students) and 92.5 (experts), which are considered excellent on the adjective rating [4].

Attention support. Features such as the Pomodoro timer and progress tracker were praised for supporting focus. S4 shared, "Some days I struggle with focusing even for 25 minutes, so being able to adjust it based on how I feel that day is great." S3 added: "It was very satisfying to see the progress bar reach a hundred percent." Experts highlighted the timer's role in promoting structured study habits. Attention support was rated between 4 and 5, with an average score

of 4.43 out of 5, suggesting that students found the tool helpful in maintaining concentration and managing fatigue.

Comprehension. Students highlighted features such as chapter summaries, quizzes, and slide annotation to improve understanding. S1 noted, "The chapter summary is helpful because sometimes you drift off during the lecture. Having the summary at the end serves as a good reminder." S2 found quizzes useful, stating "Having quizzes as I'm going through the lecture is very helpful. It makes it much easier to break things up rather than me losing attention." Comprehension averaged 4.29 out of 5, indicating that the tool effectively supported understanding.

Information retention. Participants felt that interactive revision tools, particularly mind map and quizzes, supported information retention. S7 stated, "The mind map is definitely a feature I would use. It makes everything make more sense, especially when there are lines and lines of text. It's nice to have a visual way of laying things out". S4 described the quiz feature as being similar to playing a game, saying "I love the quiz - it's almost like a game. Instead of just sitting, listening, or taking notes, you're actively doing something and testing your knowledge". Retention was rated between 3 and 5, with an average of 3.86 out of 5. The slightly lower score may be due to the short duration of the evaluation.

Accessibility. Students valued features that improved comfort and inclusivity. S2 highlighted dark mode for late-night study sessions, stating, "I prefer dark mode because I often work late at night, and the white screen can be very glaring and hard on the eyes." Subtitles and alternative formats also supported accessibility, with S2 noting, "I always end up reading captions. They help me follow along better." The text-to-speech (TTS) feature was praised for easing reading, especially when tired or due to visual stress, dyslexia, and ADHD, "I really like TTS... It makes reading much easier."

5 Conclusion

This study highlights the importance of designing educational technologies that are accessible and inclusive for HE students with ADHD. This prototype has been designed to directly address the main challenges faced by these students, including difficulty staying focused, processing unstructured content, and retaining information during video-based learning. By including features such as segmented videos, chapter summaries, interactive quizzes, and a Pomodoro timer, the tool helped break content into manageable parts and encouraged active re-engagement. The participants perceived the tool as promising in supporting HE students with ADHD to stay focused and better understand the material. The findings highlight how inclusive design can shift the focus from retrofitting accessibility to embedding support structures at the heart of educational tools, offering a more equitable experience for all learners.

6 Future Work

While the prototype showed strong initial promise, several limitations suggest directions for future research. First, AI features, while functional, relied on a basic GPT model, which may have limited the quality and specificity of generated summaries, quizzes, and chatbot responses. Future iterations will integrate validated educational content, ensuring pedagogical accuracy even when

modified by AI. Second, the tool was evaluated in a controlled setting with a relatively small sample size. Conducting longitudinal studies in real academic environments, with a more diverse participant group, would help validate its long-term effectiveness and broader applicability.

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